

Unbalance & Critical Speed Problem

A shaft of negligible mass is simply supported at both ends and carries a single disc of mass **8 kg** at its mid-span. The disc has a small residual unbalance such that its centre of mass is **0.5 mm** from the geometric axis of the shaft. The static deflection of the shaft at mid-span under the disc's weight is **1.2 mm**. The shaft runs at **1 400 rpm**.

1. Calculate the **critical speed** of the shaft.
2. Calculate the **out-of-balance force** at the running speed.
3. Determine whether the shaft is running **sub- or supercritically**.

Given Data

Symbol	Description	Value
m	Disc mass	8 kg
e	Eccentricity (CG offset)	0.5 mm = 0.0005 m
δ_{st}	Static deflection at mid-span	1.2 mm = 0.0012 m
N	Running speed	1 400 rpm
g	Gravitational acceleration	9.81 m/s ²

Critical Speed

Using the static-deflection formula:

$$N_c = \frac{60}{2\pi} \sqrt{\frac{g}{\delta_{st}}}$$

$$N_c = \frac{60}{2\pi} \sqrt{\frac{9.81}{0.0012}}$$

$$N_c = \frac{60}{2\pi} \sqrt{8175} = \frac{60}{2\pi} \times 90.42$$

$$\boxed{N_c \approx 864 \text{ rpm}}$$

Out-of-Balance Force at Running Speed

Angular velocity at running speed

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 1400}{60} = 146.61 \text{ rad/s}$$

Out-of-balance force

The centrifugal force due to the eccentric mass is:

$$F = me\omega^2 = 8 \times 0.0005 \times 146.61^2$$

$$F \approx 86.1 \text{ N}$$

Sub- or Supercritical?

$$N = 1400 \text{ rpm} > N_c = 864 \text{ rpm}$$

The shaft is running **supercritically** — above its first critical speed. In this regime the disc self-centres: the centre of mass migrates toward the bearing axis, and the dynamic behaviour stabilises. The shaft will have passed through the critical speed during run-up, where transient amplitudes must be managed by sufficient run-up rate or damping.

The speed ratio is:

$$\frac{N}{N_c} = \frac{1400}{864} \approx 1.62$$

Running at 1.62 times the critical speed places the shaft comfortably in the supercritical region. A ratio of at least 1.2–1.3 is generally recommended to ensure stable supercritical operation.

Summary

Quantity	Value
Critical speed N_c	864 rpm
Running speed N	1 400 rpm
Speed ratio N/N_c	1.62 — supercritical
Angular velocity ω	146.61 rad/s
Out-of-balance force F	86.1 N

Verification (Python)

```
import numpy as np

# Given
m      = 8.0      # kg
e      = 0.0005   # m (0.5 mm eccentricity)
delta_st = 0.0012 # m (1.2 mm static deflection)
N_run  = 1400     # rpm
g      = 9.81     # m/s^2

# Part 1 - critical speed
N_c = (60 / (2 * np.pi)) * np.sqrt(g / delta_st)
print(f"Critical speed N_c = {N_c:.1f} rpm")

# Part 2 - out-of-balance force at running speed
omega = 2 * np.pi * N_run / 60
F      = m * e * omega**2
print(f"Angular velocity    = {omega:.2f} rad/s")
print(f"Out-of-balance force F = {F:.1f} N")

# Part 3 - speed ratio
ratio = N_run / N_c
regime = "supercritical" if N_run > N_c else "subcritical"
print(f"\nSpeed ratio N/N_c    = {ratio:.2f} → {regime}")
```